Unit-II

CORROSION SCIENCE

Introduction: Most of metals (except noble metals such as Au, Pt etc.) exist in combined forms as their oxides, carbonates, hydroxy carbonates, sulphides, chlorides and silicates etc. The extraction of metals from their ores is an endothermic process, energy being supplied in the form of heat or electrical energy. Consequently, are relatively at higher energy state compared to their ores, and they have a natural tendency to revert back to their combined state. Therefore, when metals are put into use in various forms, they combine with constituents of the environment and get converted into their compounds. Thus, corrosion of metals can be considered as extractive metallurgy in reverse.

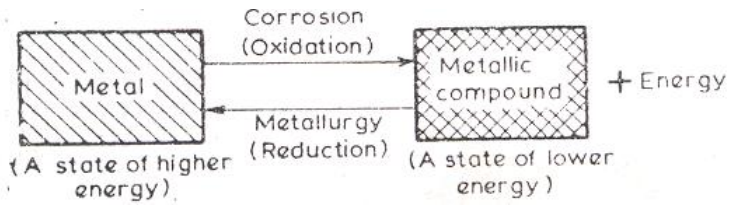


Fig. 1 Corrosion and oxidation processes

Definition: Any process of destruction (or deterioration) and consequent loss of a solid metallic material, through an unwanted chemical or electrochemical attack by its environment, starting at its surface is called corrosion.

Eg: (i) Rusting of iron: When exposed to atmospheric conditions, a layer of reddish scale (Fe_3O_4) is formed which makes the iron weak.

(ii) Formation of green film of basic carbonates ($CuCO_3 + Cu(OH)_2$) on the surface of copper, when exposed to moist-air containing carbon dioxide.

Corrosion may occur either in a dry environment or in an aqueous medium.

(i) **Dry or direct chemical corrosion:** Dry corrosion involves the direct attack of metals by dry gases mainly through chemical reactions. An important dry corrosion process is the attack of air or oxygen on the metal to form an oxide layer over the surface.

(ii) **Wet or electrochemical corrosion:** Wet corrosion, which is more common, involves reactions in aqueous medium. The conducting surface of the metal undergoes an electrochemical reaction with moisture and oxygen present in the atmosphere. This process can be explained on the basis of electrochemical theory of corrosion.

Electrochemical theory of corrosion (taking Fe as example):

When a metal, such as iron, is exposed to the environment, according to electrochemical theory, the following electrochemical changes occur gradually.

(a) Formation of a large number of minute galvanic cells (anodic and cathodic areas).

(b) Corrosion (oxidation) takes place at the **anodic area** and electrons are liberated.

$$M \rightarrow M^{n+} + ne^-$$

Eg. When iron undergoes corrosion, $Fe \rightarrow Fe^{2+} + 2e^-$

(c) The electrons released at the anodic area are taken up by the cathodic area constituting corrosion current. Consumption of electrons occurs either by (i) evolution of hydrogen or (ii) absorption of oxygen depending on the nature of corrosive environment.

(i) Evolution of hydrogen takes place in the absence of oxygen.

(a) In acidic medium and in the absence of oxygen, the cathodic reaction is,

$$2H^+ + 2e^- \rightarrow H_2\uparrow$$

(b) In neutral or alkaline medium and in the absence of oxygen, hydroxide ions are formed with simultaneous liberation of hydrogen

$$2H_2O + 2e^- \rightarrow 2OH^- + H_2\uparrow$$

(ii) Absorption of oxygen takes place in the presence of oxygen

(a) In acidic medium and in the presence of oxygen

$$2H^+ + ½\ O_2 + 2e^- \rightarrow H_2O$$

(b) In neutral or alkaline medium and in the presence of oxygen, hydroxide ions are formed by the following reaction

$$H_2O + ½\ O_2 + 2e^- \rightarrow 2OH^-$$

Corrosion of iron produces Fe^{2+} ions and OH^- ions at the anode and cathode sites respectively. These ions diffuse towards each other. Since smaller Fe^{2+} ions diffuse more rapidly than OH^- ions, their combination occurs more commonly near the cathodic region to produce insoluble $Fe(OH)_2$.

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$$

In an oxidizing environment, it is oxidized to ferric oxide and the yellow rust is hydrated ferric oxide (corrosion product).

$$2Fe(OH)_2 + ½\ O_2 + H_2O \rightarrow Fe_2O_3.3H_2O$$

In the presence of limited oxygen, ferrous hydroxide is converted into magnetic oxide of iron (F_3O_4) and is known as black rust (corrosion product).

$$3Fe(OH)_2 + ½\ O_2 \rightarrow Fe_3O_4.3H_2O$$

Types of corrosion

(a) Differential metal corrosion (Galvanic corrosion):

When two dissimilar metals are in contact with each other in a corrosive conducting medium, the metal with lower electrode potential or more active metal (higher in electrochemical series) acts as anode and with higher electrode potential (lower in electrochemical series) acts as cathode. The potential difference between two metals is the cause or driving force for corrosion. The anodic metal undergoes corrosion and the cathodic metal generally remains unattacked.

Mechanism: *At anode*, $M \rightarrow M^{n+} + ne^-$ (less noble metal)

At cathode, In acidic medium the corrosion occurs by the hydrogen evolution process; while in neutral or slightly alkaline medium, oxygen absorption occurs.

$$\text{(a)}\ 2H^+ + 2e^- \rightarrow H_2\uparrow$$

$$\text{(b)}\ H_2O + ½\ O_2 + 2e^- \rightarrow 2OH^-$$

Examples: (i) Steel screws in brass marine hardware (ii) Lead-antimony solder around copper wire (iii) Steel pipe connected to copper plumbing.

Galvanic series: In the electrochemical series (standard reduction potential arranged in an increasing order), metal high in the series is more anodic and undergoes corrosion faster than the metal below it.

For example, Li corrodes faster than Mg; Zn corrodes faster than Fe; Fe corrodes faster than Sn; Cu corrodes faster than Ag and so on. However, some exceptions to this generalization are known. For example, Ti (above Ag in the electrochemical series) is less reactive than Ag. In Zn-Al couple, Zn (below Al in the electrochemical series) is corroded; while Al acts as cathodic and is protected. These observations, exactly opposite to that predicted by the emf series, are due to the fact that metals like Ti and Al develop, strongly adhering oxide layers on their surfaces, thereby making their effective electrode potential more positive (or less negative).

From the above, it is clear that electrochemical series does not account for the corrosion of all metals and alloys. Consequently, more practical series, called *galvanic series* have been prepared by studying the corrosion of metals and alloys in a given environment like sea-water. *An arrangement of metals and alloys in the increasing order of their corrosion resistance in the given environment is referred to as galvanic series (Table-1).*

Table-1 ***Galvanic series***

1.	Mg	**Active** (or anodic)
2.	Mg alloys	
3.	Zn	
4.	Al	
5.	Cd	
6.	Al alloys	
7.	Mild steel	
8.	Cast iron	
9.	High Ni cast iron	
10.	Pb-Sn solder	
11.	Pb	
12.	Sn	
13.	Iconel	
14.	Ni-Mo-Fe alloys	
15.	Brasses	
16.	Monel (7=Ni, 30= Cu, rest=iron)	
17.	Siver solder	
18.	Cu	
19.	Ni	
20.	Cr stainless steel	
21.	18-8 Stainless steel	
22.	18-8 Mo Stainless steel	
23.	Ag	
24.	Ti	
25.	Graphite	
26.	Au	**Noble** (or cathodic)
27.	Pt	

(b) Differential aeration corrosion: Differential aeration corrosion occurs when a metal surface is exposed to differential air concentrations or oxygen concentrations. The part of the metal exposed to higher oxygen concentration acts as cathodic region and part of the metal exposed lower oxygen concentration acts as anodic region. Consequently, poorly oxygenated region undergoes corrosion.

At the anode (less O_2 concentration), $M \rightarrow M^{n+} + ne$
At the cathode (more O_2 concentration), $H_2O + ½ O_2 + 2e^- \rightarrow 2OH^-$

Thus, corrosion of metals arising as a result of the formation of an oxygen concentration cell due to the uneven supply of air on the metal surface is known as differential aeration corrosion. The following are its two types:

***(i) Waterline corrosion**:* Waterline corrosion is a case of differential aeration corrosion, more prevalent in cases such as ships, water storage steel tanks, etc., in which a portion of the metal is always under water.

The waterline corrosion takes place due to the formation of differential oxygen concentration cells. The part of the metal below the water line is exposed only to dissolved oxygen while the part above the water is exposed to higher oxygen concentration of the atmosphere. Thus, part of the metal below the water acts as anode and undergoes corrosion and part above the waterline is free from corrosion. A distinct brown line is formed just below the water line due to the deposition of rust.

Fig. 2 Water line corrosion

(ii) Pitting corrosion: The pitting corrosion is generally initiated by the deposition of sand, scale, water drop, dust etc. or due to the breakdown of the protective film. The metal below the deposit is exposed to lower oxygen concentration, acts as anode and undergoes corrosion. The metal surrounding the deposit acts as cathode, since it is exposed to higher concentration of oxygen. Because of small anodic area and large cathodic area, accelerated corrosion takes place below the deposit. Once corrosion product is formed, it further provides the condition for differential aeration below the corrosion product and surrounding metal parts. The pit grows and ultimately may case failure of the metal.

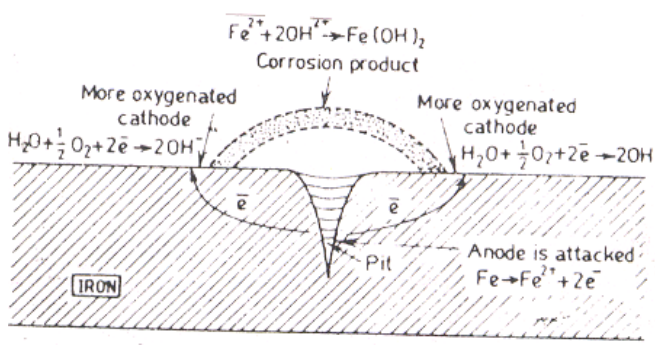


Fig. 3 Pitting corrosion at the surface of iron

Differential aeration corrosion is also observed in

1) Less aerated areas in a machinery;
2) Cracks which are not well aerated in metals;
3) Points of contact which are less aerated in a wire screen;
4) Bottom of an oil tank or bend of an oil pipe where water is trapped. Oil readily dissolves the oxygen in water creating differential oxygen concentration.

Factors affecting rate of corrosion:

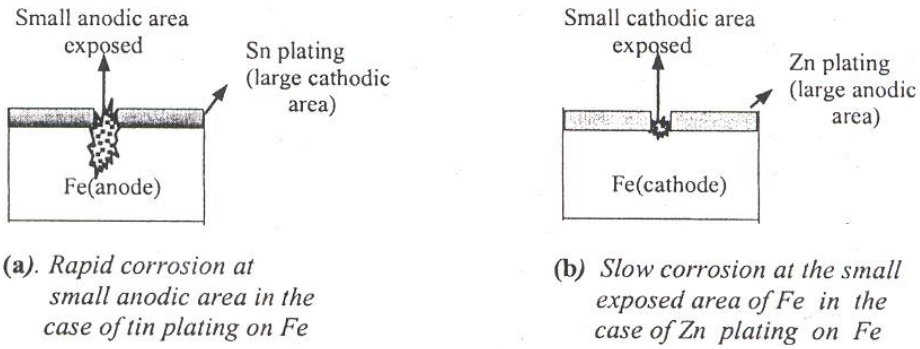
The rate and extent of corrosion, depends the following factors:

1. Related to nature of the metal:

(a) Electrode potential: When two different metals exposed to similar atmosphere are in contact with each other (galvanic corrosion), the potential difference between the anodic and the cathodic areas control the corrosion rate. Such a potential difference is referred to as the open circuit potential difference. Larger the difference in open circuit potential, higher is the rate of corrosion. For instance, the potential difference between iron and copper is 0.78V which is more than that between iron and tin (0.3V). Therefore, iron corrodes faster when in contact with copper than that with tin. Thus, farther the positions of the metals in the galvanic series, greater is the intensity of corrosion on the anodic metal. On this account, the use of dissimilar metals should be avoided wherever possible. For example, bolt and nut or screw and washer should made of the same metal (or alloy) to avoid galvanic corrosion.

(b) Relative areas of the anodic and cathodic parts: When two dissimilar metals are in contact, the corrosion of the anodic part is directly proportional to the ratio of areas of cathodic part and the anodic part. Corrosion is more rapid and severe, and highly localized, if the anodic area is small and cathodic area is large (e.g., a small steel pipe fitted in a large copper tank), because the current density at smaller anodic area is much greater, and the demand for electrons (by large cathodic areas) can be met by smaller anodic areas by undergoing corrosion more briskly.

Tin, for instance, is cathodic to iron as seen from its position in electrochemical series. If during plating of tin on iron, some areas are not covered or some pin holes are left, there results a small anode and large cathode (Fig. 5a). An intense localized corrosion occurs at the exposed small anodic surface. On the other hand, zinc plating on iron gives an anodic coating to iron since zinc is above Fe in the electrochemical series. Even if zinc plating peels off at some points intense corrosion of iron would not occur. This is because of formation of large anodic and small cathodic areas (Fig. 5b).



(a). *Rapid corrosion at small anodic area in the case of tin plating on Fe*

(b) *Slow corrosion at the small exposed area of Fe in the case of Zn plating on Fe*

Fig. 5 Corrosion rate of iron at anodic and cathodic areas.

(c) Nature of corrosion product and tendency to form protective film: The corrosion product formed on the surface of the metal may or may not act as a protective film. If the corrosion product deposited is insoluble, stable, uniform and nonporous, it acts as a protective film preventing further corrosion of metal. A thin, invisible, impervious, continuous film formed on the surface acts as a barrier between the fresh metal and corrosion environment. On the other hand, if the corrosion product is soluble, unstable, non-uniform and porous, the fresh metal surface is continuously exposed to the corrosion environment and corrosion of the metal surface takes place continuously.

In oxidizing environments, metals like aluminium, chromium, titanium, etc., are highly passive as their oxides as corrosion products form protective films on the metal surface, preventing further corrosion. Metals such as iron, zinc, magnesium etc., do not form any protective film and are highly susceptible for continuous corrosion, when exposed to oxidizing environments.

(d) Hydrogen overvoltage: Metal with low hydrogen overvoltage on its surface is more susceptible for corrosion, when cathodic reaction is hydrogen evolution type. With low hydrogen overvoltage, hydrogen gas is liberated easily and thus the cathodic reaction rate is faster. This will make the anodic reaction also faster, thereby promoting overall corrosion reaction. When the hydrogen overvoltage on the metal surface is high, cathodic reaction is slower and the corrosion of the metal also becomes slower.

2. Related to nature of environment:

(a) pH of the medium: In general, higher the acidity of the environment (low pH) higher is the rate of corrosion. At pH greater than 10, corrosion of iron practically ceases due to the formation of a protective coating of hydroxides of iron. Between pH 10 and 3, presence of oxygen is essential for corrosion of iron. A case in point of this phenomenon is that ships submerged for hundreds of years remains almost unaffected. However, if the pH is lower than 3, severe corrosion occurs even in the absence of air due to the continuous evolution of hydrogen at the cathodic region.

(b) Temperature: In general, as the temperature is increased, the corrosion is accelerated. Increase in temperature results in an increase in the conductance of the aqueous medium and hence an increase in the diffusion rate. As a consequence, corrosion progresses faster at higher temperatures. In some cases, rise in temperature decreases passivity, which again leads to an increase in the corrosion rate.

(c) Humidity: Humidity is the deciding factor in atmospheric corrosion. Critical humidity is defined as the relative humidity above which the atmospheric corrosion rate of metal increases sharply. The value of critical humidity depends on the physical characteristics of the metal as well as nature of the corrosion products. The reason why corrosion of a metal becomes faster in humid atmosphere is that gases (CO_2, O_2, etc.) and vapours, present in atmosphere furnish water to the electrolyte, essential for setting up an electro-chemical corrosion cell. Moreover, the oxide film on a metal surface, although a solid body, yet possesses the property of absorbing moisture.

(d) Presence of impurities in atmosphere: Atmosphere, in the vicinity of industrial areas, contains corrosive gases like CO_2, H_2S, SO_2, and fumes of HCl, H_2SO_4, etc. In presence of these gases, the acidity of the liquid, adjacent to the metal surfaces, increases. This consequently results in an increase of corrosion current, flowing in the local electro-chemical cells on the exposed metal surfaces. Similarly, in the marine atmosphere, the presence of sodium and other chlorides (of sea water) leads to increased conductivity of the liquid layer in contact with the metal surface, thereby corrosion is speeded up.

Corrosion control:
Protective coatings

(1) Inorganic coatings: Inorganic coatings are generally chemical conversion coatings. A surface layer of the metal is converted into a compound, by chemical or electrochemical reactions, which forms a barrier between the underlying metal surface and corrosion environment. Such coatings are particularly used as excellent base for paints, lacquers oils and enamels. An example of inorganic coating is Anodizing.

Anodizing (Anodized coating): Anodized coating is generally produced on non-ferrous metals like Al, Zn, Mg and their alloys by anodic oxidation process, in which the base metal is made as anode. It is carried out by passing a moderate direct current through a bath in which the metal or alloy is suspended from anode. The bath usually contains sulphuric acid, chromic acid, phosphoric acid, oxalic acid or boric acid.

Aluminium coatings are formed by electrolysis in a suitable acid electrolyte (sulphuric, chromic, oxalic or phosphoric acid) at moderate temperature (35-40^{0}C) and moderate current densities (10-20 mAcm^{-2}). The coating forms as a result of progressive oxidation, starting at the aluminum surface. An oxide film, initially very thin, grows from the metal surface outwards and increases in thickness as oxidation continuous at aluminium anode. The outer part of the oxide film formed is very porous and considerably softer than the layer directly adjoining the metal surface. The minute exposed pores can be made more corrosion-resistant by the process of sealing, which consists of exposing film to boiling water. This treatment porous alumina at the surface of coating into its monohydrate ($Al_2O_3.H_2O$), which occupies more volume, there-by the pores are sealed. Anodized articles are used as soap boxes, tiffin carriers, window frames, etc.

(2) Metal coatings: Deposition of a protective metal over the surface of a base metal is called metal coating. Metal coatings are applied by electrodeposition, flame spraying, cladding, hot dipping, diffusion coating and vapor deposition.

(i) Anodic metal coating: Anodic coatings are produced by coating a base metal with more active metals which are anodic to the base metal. For example, iron is coated with anodic and more active metals such as Zn and Mg. One of the important characteristics of anodic coating is that, even if the coating is ruptured, the base metal does not undergo corrosion. The exposed surface of the metal is cathodic with respect to the coating metal and the coating metal preferentially undergoes corrosion. The protection is ensured as long as the anodic coating metal is still present on the surface. Therefore, anodic coating is also known as *sacrificial coating*. Galvanization is a familiar example for anodic coating and is extensively used to protect iron and steel objects.

Galvanizing (Galvanization) is a process of coating a base metal surface with zinc metal. Galvanization is carried out by hot dipping method. The galvanization process involves the following steps.

1. The metal surface is first washed with organic solvents to remove oil and grease deposits.
2. Rust and other deposits are removed by washing with dilute sulphuric acid (pickling).
3. Finally, the article is washed with water and air dried.
4. The article is then dipped in a bath of molten zinc, maintained at 425-430^{0}C and covered with a flux of ammonium chloride to prevent the oxidation of molten zinc.
5. The superfluous (excess) zinc on the surface is removed by passing through a pair of hot rollers, which wipes out excess of zinc coating and produces a thin coating.

Uses: It is most widely used for protection of iron from atmospheric corrosion in the form of roofing sheets, wires, pipes, nails, bolts, screws, buckets, tubes etc. It may be pointed here that zinc gets dissolved in dilute acids, to form highly toxic (or poisonous) compounds. Hence, galvanized utensils cannot be used for preparation and storing foodstuffs, especially acidic ones.

(ii) Cathodic metal coating: Cathodic coatings are produced by coating a base metal with a more noble metal which is cathodic. The metals such as Cu, Ni, Sn, Ag are commonly used as cathodic coatings on steel. These coating metals are less reactive than the base metal and are less susceptible for corrosion. The cathodic coating provides protection only when it is undamaged, continuous and non-porous. When the coating is discontinuous, an intense localized corrosion occurs due to the setting up of a galvanic cell that consists of a large cathodic area and small anodic area.

Tinning is a process of coating a base metal with tin. Tinning of iron or steel is carried out by hot dipping method and the process involves the following steps.

1. The metal surface is first washed with organic solvents to remove oil and grease deposits.
2. Rust and other deposits are removed by washing with dilute sulphuric acid (pickling).
3. Finally, the article is washed with water and air dried.
4. The clean and dry sheet is passed through molten zinc chloride flux. The flux helps in molten metal to adhere on the metal surface.
5. It is then passed through a tank that contains molten tin.
6. Finally, it is passed through a series of rollers immersed in palm oil. The rollers wipe out excess tin deposit and produce a continuous, thin coating of tin on the sheet. The oil prevents the oxidation of the tin coated surface.

Uses: Because of non-toxic nature of tin, tinning is widely used for coating steel, copper, and brass sheets, used for manufacturing containers of storing foodstuffs, ghee, oils, kerosene and packing food materials. Tinned-copper sheets are employed for making cooking utensils and refrigeration equipments.

(3) Cathodic protection: Cathodic protection is method of protecting a metal or alloy from corrosion by converting it completely cathodic and no part of it is allowed to act as anode. As known, the corrosion of metal takes place at anodic region whereas at the cathodic region metal is unaffected. Therefore, corrosion can be prevented by eliminating the anodic sites and converting entire metal into cathodic area. Cathodic protection can be achieved by following sacrificial anode method.

Sacrificial anode method: In this protection method, the metallic structure (to be protected) is connected by a wire to a more anodic metal, so that all the corrosion is concentrated at this more active metal. The more active metal itself gets corroded slowly; while the parent structure (cathodic) is protected. The more active so-employed is called sacrificial anode. The corroded sacrificial anode block is replaced by a fresh one, when consumed completely. Metals commonly employed as sacrificial anodes are Mg, Zn, Al and their alloys. Important applications of sacrificial anodic method include protection of buried pipelines, underground cables, marine structures, ship-hulls, water-tanks, piers, etc. (Fig. 4).

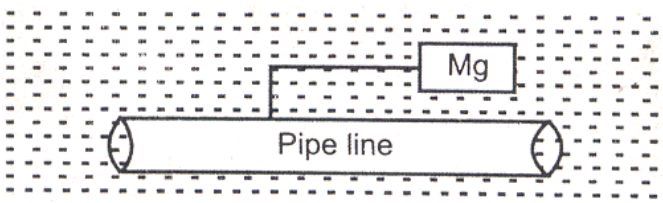


Fig. 4 Cathodic protection by sacrificial anode method

Corrosion Penetration Rate (CPR)

The corrosion penetration rate (CPR) is defined as the speed at which any metal in a specific environment deteriorates due to a chemical reaction in the metal when it is exposed to a corrosive environment or the amount of corrosion loss per year in thickness.

The speed or rate of deterioration depends on the environmental conditions and the type and condition of the metal under study. Corrosion penetration rate may also be known as corrosion rate.

Weight-loss method

Weight-loss tests are mostly used for measurement of corrosion. In this, cleaned metal coupon or a standard test piece is measured, weighed, and exposed to the corrosive medium for a known time. The piece is then taken out, cleaned to remove the corrosion products and reweighed.

The rate of corrosion of the metal, CPR is then calculated from the equation

$$\text{CPR} = {}^{KW}/_{ATD}$$

where K is constant, T is the time of exposure (in hours), W is the weight loss (in mg), A is the exposed surface area of the metal and D is density of the metal (in g/cm^3).

The result is commonly expressed in mils per year (mpy) (1 mil = 1/1000 inch) or millimeters per year (mmpy).

Note: When CPR is calculated in mpy, $K = 534$ and metal surface area is expressed in inch^2. When CPR is calculated in mmpy, $K = 87.6$ and metal surface area is expressed in cm^2.

Numerical

1. Calculate the CPR in both mpy and mm/year for a thick steel sheet of area 100 inch^2 which experiences weight loss of 485 g after one year. The density of steel is 7.9 g/cm^3.

Solution: To find CPR in mpy

$$\mathbf{CPR} = \frac{\mathbf{k \times W}}{\mathbf{D \times A \times T}}$$

$$\mathbf{CPR} = \frac{534 \times 485 \times 1000}{7.9 \times 100 \times 365 \times 24} = \mathbf{37.42\ mpy}$$

To find CPR in mm/year:

$$\mathbf{CPR} = \frac{\mathbf{k \times W}}{\mathbf{D \times A \times T}}$$

$$\mathbf{CPR} = \frac{87.6 \times 485 \times 1000}{7.9 \times 100 \times 6.45 \times 365 \times 24} = \mathbf{0.9518\ mm/year}$$

METAL FINISHING

Metal finishing and its technological importance:

Definition: The term metal finishing applies to all surface processes which are carried out in order to change the metal surface properties by deposition of layer of another metal or a polymer or by the formation of an oxide film.

The important metal finishing techniques are

(i) Electroplating of metals, alloys and composites.
(ii) Electrolessplating of metals, alloys and composites.
(iii) Immersion plating of metals (involves chemical displacement of metals).
(iv) Chemical conversion coatings based on metal compounds (involves the deposition of inorganic coatings such as oxides, chromates, phosphates and their mixtures. E.g., anodizing, phosphating, chromating).
(v) Electrophoretic painting which involves painting on electrically charged conducting surface such as motor car body.

Electroplating

Definition: Electroplating is the deposition of metal, by electrolysis, over the surface of a substrate; the latter may be another metal, polymer, ceramic or a composite. Alloy or metal-polymer or metal-composite deposition is also carried out by electroplating.

The following types of electrodeposition are in practice.

(i) Single metals: the most important are Sn, Cu, Ni, Cr, Zn, Cr, Pb, Ag, Au and Pt.
(ii) Alloys: including Cu-Zn, Cu-Sn, Pb-Sn, Sn-Ni, Ni-Co, Ni-Cr and Ni-Fe.
(iii) Composites: metals containing dispersed solids such as PTFE (Teflon), Al_2O_3 WC, diamond, SiC, Cr_3C_2 and graphite.

Principle of electrodeposition:

The electrodeposition is carried out by electrolyzing a suitable salt solution of the metal being plated. The substrate is the cathode and the anode is usually the pure metal. On electrolysis, the anode dissolves into the electrolytic bath and the metal ions from the bath solution are deposited as metal over the cathode. The reactions taking place for an electrolyte MA are given below:

The electrolyte (MA) ionizes in aqueous solution

$$MA\ (aq) \rightleftharpoons M^{n+}(aq) + A^{n-}(aq)$$

At anode, the electrode dissolves giving raise to metal ions

$$M \rightarrow M^{n+} + ne^- \ldots\ldots\ldots(1)$$

At cathode, the metal ions get reduced and are deposited as metal atoms

$$M^{n+} + ne^- \rightarrow M \ldots\ldots.(2)$$

In an ideal situation, the conditions of electrolysis are controlled in such a way that the current efficiency of the reactions (1) and (2) are equal so that the concentration of M^{n+} remains the same in the electrolyte.

In some cases, where an inert anode is used, the main anode reaction is oxygen evolution and hence metal ions are added in the form of metal salt to the electrolyte to maintain the concentration.

$$H_2O \rightarrow \frac{1}{2} O_2\uparrow + 2H^+ + 2e^-$$

In acid conditions, H_2 may get liberated at the cathode as a side reaction.

$$H^+ + e^- \rightarrow \frac{1}{2} H_2$$

It is necessary to have a highly conducting electrolytic solution and current density ($mAcm^{-2}$) is adjusted so as to get a good deposit.

Mechanism of electrodeposition: Deposition of metal is similar to crystal growth since metals have well defined crystal structures. Electrodepositon takes place in two distinct phases. In the first phase, formation of nuclei takes place covering the electrode with a few atomic layers of the metal. In the second phase, growth of the deposited resulting in the thickening of the layer into a macroscopic (1-10 μm thick) deposit takes place. Formation of nuclei requires a high overvoltage. However, nuclei once formed will grow quite rapidly at a relatively low overvoltage. The atoms (adatoms) that are formed during the plating process on the crystal plane quickly move and occupy favourable sites. The most favourable site is kink site where the atom can interact with three neighbours. Atoms also occupy less favourable sites such as edge site (two neighbours) or just remain as adatoms (one neighbour). However, they tend to diffuse to more favourable sites. The overall phase growth takes place in the following sequence:

(a) Mass transport in solution by diffusion of ions to the electrode surface
(b) Electron transfer to form adatom
(c) Surface diffusion i.e., the diffusion of the adatoms across the surface into the most favourable positions (kink sites)

A good electrodeposit is characterized by good adhesion, fine-grained texture, uniformity in thickness, good covering power and brightness.
A poor deposit has macroscale features such as steps, ridges and polycrystalline block growth resulting in a deposit which is coarsely crystalline, nonuniform and dull in appearance. It is characterized by an outward growth often resulting in a deposit with powdery (burnt) texture.
During plating operation, the factors are so adjusted that steps (b) and (c) predominate.

Effect of plating variables on the nature of electrodeposit (Factors governing electroplating):

(1) Current density of deposition: The current density (c.d.) of deposition is defined as the current per unit area generally expressed in milliamperes per square centimeter (mA/cm^2) of the electrode surface. As the applied voltage is increased the current density rises, till it reaches a limiting value.
At low current densities, surface diffusion is fast compared with electron transfer and the adatoms finds the most favourable position (kink sites) resulting in a well-formed deposit. As the current density is increased, surface diffusion is slowed down compared to the electron transfer and the adatoms may not reach the most favourable positions. Further, at high potentials, number of nuclei formed may also increase. These may lead to a less ordered deposit with macroscale features.
At very high current densities (below limiting current region) mass transport predominates in solution resulting in a bad deposit with rough and powdery texture which is poorly adherent to surface.

At very high current densities above the limiting current, hydrogen evolution on the surface predominates causing a fast depletion of H^+ ions in the vicinity of the cathode. This results in dispersed powders containing oxides and hydroxides of the metal.
In practice, *optimum current density* for a particular deposition is actually determined and used so as to get a good deposit.

(2) Plating bath: The bath contains the solution used for plating. It is normally a mixture of metal ion solution, other electrolytes, complexing agents and various organic additive added to improve the nature of the deposit.

(a) Metal ion concentration and electrolytes: Normally, metal ion concentration (either as simple or complex ion) is moderate (about 1-3 mole/dm^3). By employing still higher metal-ion concentration, the mass transfer process decreases, thereby badly affecting the quality of the deposit. Moderate metal ion concentration, however, yields very adherent coating films. For getting thick deposit of metal, somewhat higher metal ion concentration is used.
Electrolytes are also added in high concentrations to increase the conductivity of the plating bath. Electrolytes do not participate in the electrode reactions but only increase the conductivity of the plating and sometimes exert buffer action and control the variation of pH.

(b) Complexing agents: The metal ion is converted into a complex ion by the addition of suitable complexing agents so as to get a fine-grained and more adherent deposits than that obtained from simple ion solutions. Complex ions are used

(i) To prevent the chemical reaction between the cathode metal and plating ions (e.g., in plating Cu on Fe or steel, copper ion is complexed to prevent the reaction $Cu^{2+} + Fe \rightarrow Cu + Fe^{2+}$).
(ii) To make the potential of the plating ion more negative so that the plating takes place at lower potential.
(iii) To prevent the passivation of anodes so that anodes dissolve easily and thus increase the current efficiency (e.g., Chloride ions in low concentration is used in many baths).
(iv) To improving the throwing power of the plating bath.

The most frequently used complexing agents are cyanides, hydroxides and sulphamates. Cu, Ag, Au, Zn and Cd are plated as smooth deposits and with maximum efficiency in the presence of cyanide ions.

(c) Organic additives: Certain organic compounds added to the plating baths have a remarkable influence on the nature of the electrodeposit. They modify the structure, morphology and properties of the electrodeposit. These include brighteners, levelers, structure modifiers and wetting agents.
(i) Brighteners: For a deposit to be bright, the light falling on the surface should be reflected and not scattered. This happens when the grain size of the deposit is lower than the wavelength of the incident light. Brighteners are added to produce microscopically fine deposits. These plate over the metallic surface and probably function by encouraging the plating to the substrate surface and inhibiting the plating perpendicular to the surface. Examples: Aromatic sulphones or sulphonates and molecules containing C≡N, N=C=S, C=O groups (e.g., thiourea and coumarin).
(ii) Levellers: Plating takes place rapidly on certain regions such as on dislocations where the diffusion of ions is faster resulting in a deposit of uneven thickness. Levellers help to produce a level deposit by getting adsorbed at regions where rapid deposition takes place. Thus, adsorbed additives reduce the rate

of electron transfer at such places. Many brighteners also act as levelers. Example: Sodium allyl sulphonate is used as a leveler for nickel deposition.

(iii) Structure modifiers (Stress relievers): Structure modifiers modify the structure of the deposit in such a way as to alter the deposit properties. For example, all electrodeposits are associated with internal stress (due to lattice misfit) and may give away to impact if the stress levels are too high. Sometimes this results in microcracking of the deposit. Example: Saccharin.

(iv) Wetting agents: During the process of electroplating, hydrogen gas is evolved on the cathode. Wetting agents are added to release the gas bubbles from the surface. They also improve the leveling and uniformity of the deposit and reduce the deposit brittleness. In the absence of wetting agents, the gas bubbles may get entrapped within the plated metal and try to escape after the plating process. This often makes the substrate metal itself brittle when subsequently heat treated (hydrogen embrittlement). Example: Sodium lauryl sulphate.

(3) pH: At low pH values, evolution of hydrogen occurs on the cathode, resulting in a burnt deposit. In such cases, pH has to be increased. On the other hand, at higher pH values the electrode surface gets coated with insoluble hydroxides. Therefore, the optimum pH range for most of the plating processes is from 4 to 8, i.e., slightly acid or slightly alkaline. The desired pH in the plating process is maintained using buffers. Example: Borate buffer is used in nickel plating and citrate buffer in gold plating.

(4) Temperature: A good deposit is formed at slightly elevated temperatures because of the increase in the surface diffusion of ions. However, high temperatures may lead to hydrogen evolution on the cathode, corrosion of process equipment and sometimes decomposition of the organic additives. At moderate temperatures, surface diffusion is more than the hydrogen evolution and hence plating of many metals is carried out between 35 and 60 ^{0}C.

(5) Throwing power of the plating bath: The ability of a plating bath to give a uniform and even deposit on the entire surface of the object is measured by its throwing power. This is of particular importance in objects which have complex shapes. For example, plating in holes and recesses of the objects takes place less uniformly than on even because the former points are at a lower over potential. If the distribution of the deposit is uniform, irrespective of the shape of the object, throwing power is said to be good.

Electroplating of Chromium:

Chromium plating is a finishing treatment utilizing the electrodeposition of a thin layer of chromium on a metal substrate. This layer can either be decorative or corrosion protective to the base surface.

Pretreatment: Metallic surfaces (substrate) are normally associated with greasy materials or a thin film of metallic oxide. The substrate surface should be made free from dirt, oil and grease before electroplating.

Degreasing of substrate surface is done either by treating the surface with organic solvents or aqueous alkali treatment. This is followed by “Pickling” which involves neutralization of alkali and dissolution of oxide layer by immersion in 10% sulphuric acid. Further, polishing of the substrate is done either through mechanical grinding and buffing or anodic dissolution of the surface of the substrate known as

electropolishing. Finally, the substrate is rinsed with deionized water and dried before it is subjected to electroplating.

Electrolysis: After the pretreatment, the anode and the object are connected respectively to the positive and negative terminals of the bus bars (or jigs) and are placed in the electrolyte. The electrolyte is maintained at the desired temperature. The plating operation is conducted at the suitable voltage by maintaining the current density for the required time to get a smooth deposit of the metal on the cathode. In general, the thickness of the deposit ranges from 0.01 to 100 µm, depending up on the application. The time required for electroplating varies from a few seconds to thirty minutes. The conditions under which electroplating of Chromium are given below:

	Chromic acid bath	
	Decorative Chromium	**Hard Chromium**
Plating bath solution	Chromic acid (CrO_3) and H_2SO_4 in 100:1 proportion	
Operating temperature	45-60^{0}C	43-66 ^{0}C
Current density	100-200 mAcm^{-2}	250-430 mAcm^{-2}
Current efficiency	8-12%	10-15%
Anode	Insoluble anodes- Pb-Sb or Pb-Sn coated with PbO_2 or stainless steel	
Cathode	Object to be coated; pretreated	
Appearance	A thin coating with bright and polished surface	A thick coating with a durable and wear-resistant surface
Applications	Coating for consumer goods (car parts, kitchenwares, sanitary wares, etc.)	Coating for industrial machinery and tools (hydraulic cylinders, gears, pistons, etc.)

The plating bath contains CrO_3 in which Cr is in +6 oxidation state. This is reduced +3 oxidation state by a series of complex reactions in the presence of SO_4^{2-} (catalyst) furnished by H_2SO_4. Cr^{3+} ions are reduced to elemental chromium which gets deposited on the substrate.

$$CrO_3 + H_2O \rightarrow H_2CrO_4 \rightarrow CrO_4^{2-} + 2H^+$$

$$2H_2CrO_4 \rightarrow H_2Cr_2O_7 + H_2O \rightarrow Cr_2O_7^{2-} + 2H^+ + H_2O$$

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

$$Cr^{3+} + 3e^- \rightarrow Cr$$

The amount of Cr^{3+} ions should be restricted in order to obtain good deposit. Insoluble anodes like Pb-Sb or Pb-Sn alloys covered with a layer of PbO_2 (which oxidize Cr^{3+} to Cr^{6+}) are used thus controlling the Cr^{3+} concentration.

$$2Cr^{3+} + 3\,O_2 \rightarrow 2CrO_3 + 6e^-$$

Chromium anodes are not used in Cr plating because (i) Chromium metal passivates strongly in acid sulphate medium. (ii) Chromium anode gives rise to Cr^{3+} ions on dissolution. Therefore, in the presence of large concentration of Cr^{3+} ions, a black Cr deposit is obtained.

Electroless Plating

Definition: Electroless plating is the controlled deposition of a continuous film of a metal from its salt solution on to a catalytically active surface by a suitable reducing agent without using electrical energy.

The reducing agent brings about the reduction of the metallic ions to the metal which gets plated over a catalytic surface. The driving force in electroless plating is an autocatalytic redox reaction on a pretreated active surface.

$$\text{Metal ion + Reducing agent} \xrightarrow{\text{catalyst surface}} \text{Metal + Oxidized product}$$

An important aspect of electroless plating is the preparation of the surface of the object so that an active surface is obtained. This is achieved by one of the following techniques:

(a) Acid treatment (etching)
(b) Electroplating a thin layer of the metal. (same as that to be plated or another suitable metal) followed by heat treatment
(c) For nonconducting surfaces such as plastics or printed circuit boards, the surface is treated with stannous chloride and palladium chloride solution alternately

Advantages of electroless plating:

(i) Use of electrical power and electrical contacts are eliminated.
(ii) Semiconductors and insulators like plastics can also be plated.
(iii) Electroless baths have better throwing power. Intricate parts with irregular shapes, inside parts of tubes, recesses can be uniformly coated.
(iv) Hydrogen gas liberated is not trapped in blind holes.
(v) Electroless coating are harder than that produced by conventional electroplating.
(vi) No levelers are required.
(vii) Deposits have unique chemical, mechanical and magnetic properties.

Electroless plating of Copper (on PCB):

<u>Pretreatment and activation of the surface:</u> The substrate is degreased and etched in acid. It is activated by dipping in stannous chloride solution containing HCl at 25^0C followed by dipping in palladium chloride. The surface is dried to get a layer of palladium.

<u>Plating bath solution:</u>
A solution of copper sulphate (12 g/dm^3)
Reducing agent- Formaldehyde (8 g/dm^3)
Buffer - Sodium hydroxide (15 g/dm^3) + rochelle salt (14 g/dm^3)
Complexing agent and exaltant – EDTA (20 g/dm^3)
pH – 11 ; Temperature – 25^0C

<u>Reactions:</u> Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$

Anode: $2HCHO + 4OH^- \rightarrow 2HCOO^- + 2H_2O + H_2 + 2e^-$

Overall reaction: $Cu^{2+} + 2HCHO + 4OH^- \xrightarrow{\text{catalyst surface}} Cu + 2HCOO^- + 2H_2O + H_2$

<u>Rate of plating:</u> 1 to 5 μm h^{-1}

Application: An important application of electroless copper plating is printed circuit boards (PCB). In double side and multilayer boards where plating-through-holes is required, electroless technique is the best since the holes cannot be electroplated

The technique used to manufacture printed circuit boards is the subtractive method (Fig. 5). The base object in PCB is a plastic material such as glass fibre reinforced plastic (GRP), epoxy or phenolic polymer over which a layer (5 to 100 μm) of copper is electroplated. Selected areas are then protected by a photo-resist or electroplated image and the rest of the copper is etched away to produce the circuit pattern or the track. Usually, double sided tracks are made in order to pack more number components in a small space. Finally, the connection between the two sides of PCB is made by drilling holes, followed by electroless copper through holes.

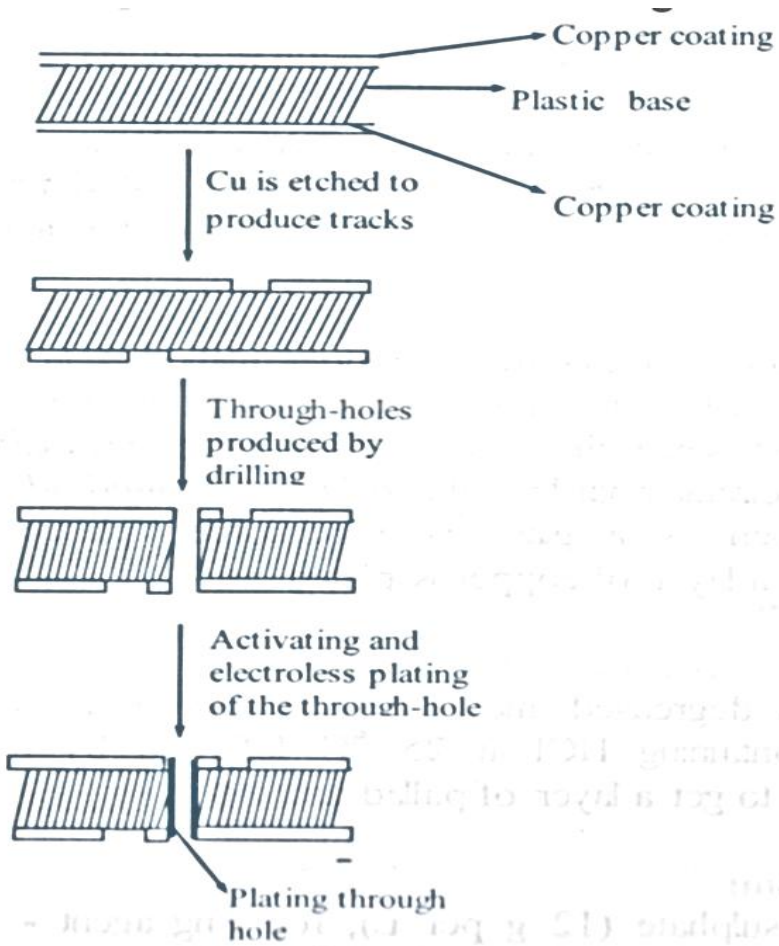


Fig. 5 Steps in subtractive technique of plating-through in the manufacture of double-sided PCB. Copper cladded tracks on each side of the plastic board are connected through-hole.

Comparison between electroplating and electroless plating

	Electroplating	***Electroless plating***
Driving force	Passage of current	Autocatlytic redox reaction
Anode	A separate anode is used	Catalytic surface of the substrate acts as anode
Anode reactant	Pure metal (M) or H_2O	Reducing agent
Anode reaction	$M \rightarrow M^{n+} + ne^-$ or $n/2\ H_2O \rightarrow n/2\ O_2 + nH^+ + ne^-$	Reducing agent → Oxidized product + ne^-
Cathode	Object treated to remove surface impurities	Object treated to make the surface catalytically active
Cathodic reaction	$M^{n+} + ne^- \rightarrow M$	$M^{n+} + ne^- \rightarrow M$
Applicability	Only to conductors	Both conductors and non-conductors
Throwing power	Not satisfactory for intricate parts with irregular shape and inside of tubes	Satisfactory irrespective of the shape and inaccessibility of the parts.
Nature of the deposit	Pure metal or alloy	Metals with reducing agents and oxidized products as impurities

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